

Yuanhao Zou

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Education

- University of Central Florida, Orlando**, PhD. in Computer Science Aug 2025 – Present
- GPA: 4.0/4.0
 - **Advisor:** Chen Chen
 - **Research Interest:** Video Understanding, Efficient Vision Language Model, Multimodal Learning.
- University of Michigan, Ann Arbor**, MS in Electrical and Computer Engineering Aug 2023 – May 2025
- GPA: 4.0/4.0
- Central South University, China**, BE in Computer Science and Technology Sep 2019 – Jun 2023
- GPA: 88/100

Main Publications

- [NeurIPS 2026 Submission, First Author] Conditional Multi-Event Temporal Grounding in Long-Form Video
- [CVPR 2026 Workshop, First Author] How Should Video LLMs Output Time? An Analysis of Efficient Temporal Grounding Paradigms
- [ECCV 2026, Second Author] CLARITY: Medical World Model for Guiding Treatment Decisions by Simulating Context-Aware Disease Trajectories in Latent Space
- [ICLR 2026, First Author] A.I.R.: Enabling Adaptive, Iterative, and Reasoning-based Frame Selection For Video Question Answering
- [CVPR 2025, First Author] Alignment, Mining and Fusion: Representation Alignment with Hard Negative Mining and Selective Knowledge Fusion for Medical Visual Question Answering
- [CVPR 2025 Workshop, First Author] MVCM: Enhancing Multi-View and Cross-Modality Alignment for Medical Visual Question Answering and Medical Image-Text Retrieval
- [Knowledge-Based Systems, First Author] HFA-UNet: Hybrid and Full Attention UNet for Thyroid Nodule Segmentation

Internship

- Axon (Part-Time)** Oct. 2025 – Dec. 2025
- **Efficient Vision Language Model:** Developing efficient video anomaly detection model for edge devices like Axon AB4, effectively handling images in low-level and extreme conditions.
- United Imaging (Full-Time)** May. 2026 – Aug. 2026
- **Multi-Modality Video Reasoning Agent using Reinforcement Learning:** Training an RL-based agent that works on Multi-Modality Videos (Thermal, Depth, and RGB) to solve complex questions that require joint reasoning ability.

Research Experience

- University of Central Florida, Prof. Chen Chen** Feb. 2025 – Present
- **Video Understanding:** Frame Selection Model integrates fast lightweight models like CLIP and heavy vision language model, achieving better performance and efficiency.
 - **Efficient Vision Language Model:** Developing efficient vision language models that used on edge devices. **Outcomes:** First Author of one CVPR26 Workshop Paper.
 - **Conditional Multi-Event Temporal Grounding:** Designing a benchmark and new method. A project collaborated with Qualcomm.

Stony Brook University, Prof. Zhaozheng Yin

Feb 2024 – Nov 2024

- **Medical Vision-Language model:** (1) Addressed challenges in cross-modality understanding and the underutilization of multi-view images in medical radiology datasets, and applied to Medical VQA and Medical Image-Report Retrieval. (2) Developed a unified representation alignment approach with hard negative mining and selective knowledge fusion to enhance Med-VQA performance significantly.
- **Outcomes:** First Author of two research papers to **CVPR 2025**.

University of Nottingham Ningbo China (UNNC), Prof. Xiangjian He

Sep 2022 – Aug 2023

- **Medical Image Segmentation:** Developed a deep learning network integrating U-Net and Transformer for hybrid attention and multi-scale fusion modules to address challenges with limited samples and small objects in a new cervical dataset.
- **Outcomes:** a paper published to Journal **Knowledge-Based Systems**.

Other Publications

- **[International Workshop on Advanced Imaging Technology (IWAIT) 2024]** Multimodality semisupervised learning for ophthalmic biomarkers detection.
- **[International Workshop on Advanced Imaging Technology (IWAIT) 2024]** Alignment, Mining and Fusion: Representation Alignment with Hard Negative Mining and Selective Knowledge Fusion for Medical Visual Question Answering.

Technologies

Proficient Technologies: Pytorch, Linux, Lightning, Git